Quiz 6

Graded

Student

Clarisse Cheung

Total Points

0 / 5 pts

Question 1

Question 1

0 / 2.5 pts

- **– 0 pts** Full Credit
- **– 0.25 pts** Majority Credit
- **– 1.25 pts** Half Credit
- ✔ **– 2.5 pts** No Credit

1. Encryption, in general, is non determinisitic unless specified otherwise
2. To solve this question, it's useful to think of the Enc algorithm as having a second input r which characterizes all of the random elements of Enc given its inputs pk, m. An unbound adversary can iterate through all of these possible outputs of Enc for the two messages and return when it finds a matching ciphertext to the input.

Question 2

Question 2

0 / 2.5 pts

- **– 0 pts** Full Credit
- **– 0.25 pts** Majority Credit
- **– 1.25 pts** Half Credit
- ✔ **– 2.5 pts** No Credit

3. Your argument needs to work for an arbitrary message space.
4. To answer this question correctly, you need to propose an adversary that conforms to the definition of IND-security, and then show that it has non-negligible advantage.
5. Here's an adversary with advantage 1: choose any two distinct messages m_0, m_1 of the same length. Encrypt $c = \mathsf{Enc}_{\mathsf{pk}}(m_0)$. On challenge ciphertext c^*, output 0 if $c = c^*$ and otherwise output 1. (To analyze this, use the law of total probability and correctness).

CMSC 28400 **Quiz 6 (2 questions)** November 3, 2026

Name: Clarisse Cheung **CNET ID:** clarissec @uchicago.edu

Please write neatly, not too small, and not too faint.

Question 1: Let Π be a correct PKE scheme for 1-bit messages. Show that, given pk and a ciphertext c computed via $c \leftarrow \mathsf{Enc}_{\mathsf{pk}}(m)$, it is possible for an unbounded adversary to determine m with probability 1.

We construct adversary A:

A(pk, c):

$c^* := Enc_{pk}(0)$

if $c^* = c$ return 0

else return 1

1. A is PPT. Since Enc is PPT and it only calls it once with string comparison, A is PPT.

2. We have 2 cases:

① $m = 0$. Since $c^* = Enc_{pk}(0)$ and Enc is deterministic, then $c^* = c$. Thus A returns 0 and is correct.

② $m = 1$. Say $c^* = Enc_{pk}(0) = Enc_{pk}(1) = c$. By correctness, this is not possible as $Pr[Dec_{sk}(Enc_{pk}(m)) = m] = 1$ so there cannot be some $c^* = c$ that has 2 different m's. Thus A returns 1 and is correct.

Question 2: Let Π be a correct PKE scheme with a deteministic encryption function Enc, over a general message space $\mathcal{M}$ with $|\mathcal{M}| > 1$. Prove that Π cannot be IND-secure.

let $M = \{0,1\}$ since an adversary should be able to pick any 2 messages and not differentiate. Let $m_0 = 0$ and $m_1 = 1$. 3

4

$Adv(n) = |Pr(\hat{b} = b) - \frac{1}{2}|$ and uniformly sampled so $Pr(m_0) = Pr(m_1) = \frac{1}{2}$

1. Let $Pr[\hat{b} = b \mid m = 1] = P_1$. Let $Pr[\hat{b} = b \mid m = 0] = P_0$
2. By LOTP, $Pr[\hat{b} = b \mid m = 1] = 1 - Pr[\hat{b} = b \mid m = 0] = 1 - P_0$.
3. $Pr(\hat{b} = b) = \frac{1}{2} \cdot P_0 + \frac{1}{2} \cdot (1 - P_0) = \frac{P_0}{2} + \frac{1}{2} - \frac{P_0}{2} =$

5

$$\left| Pr[\hat{b} = b \mid m = 1] - Pr[\hat{b} = b \mid m = 0] \right| = \frac{1}{2}$$

$\frac{1}{2}$ is not negl so π cannot be IND secure